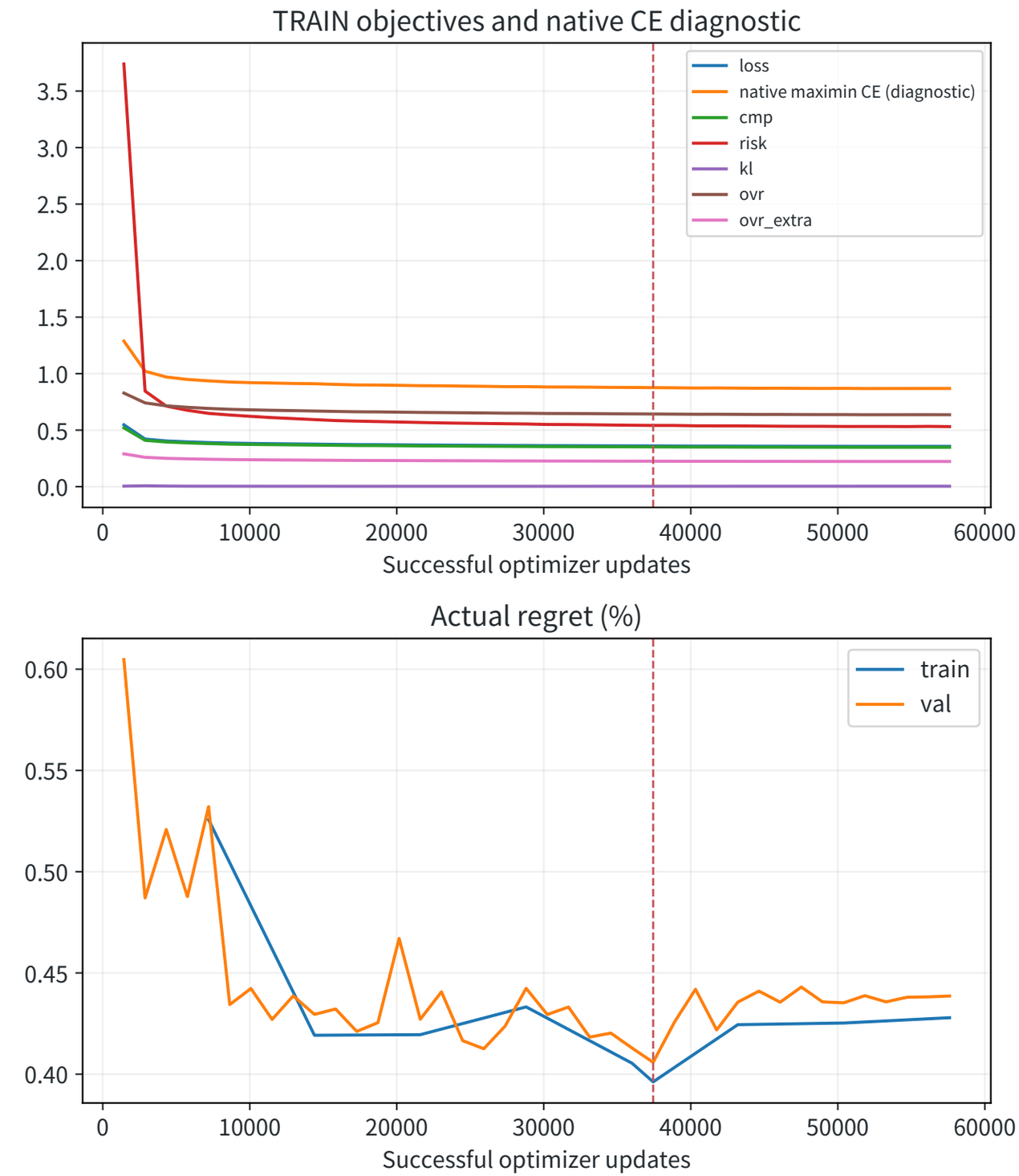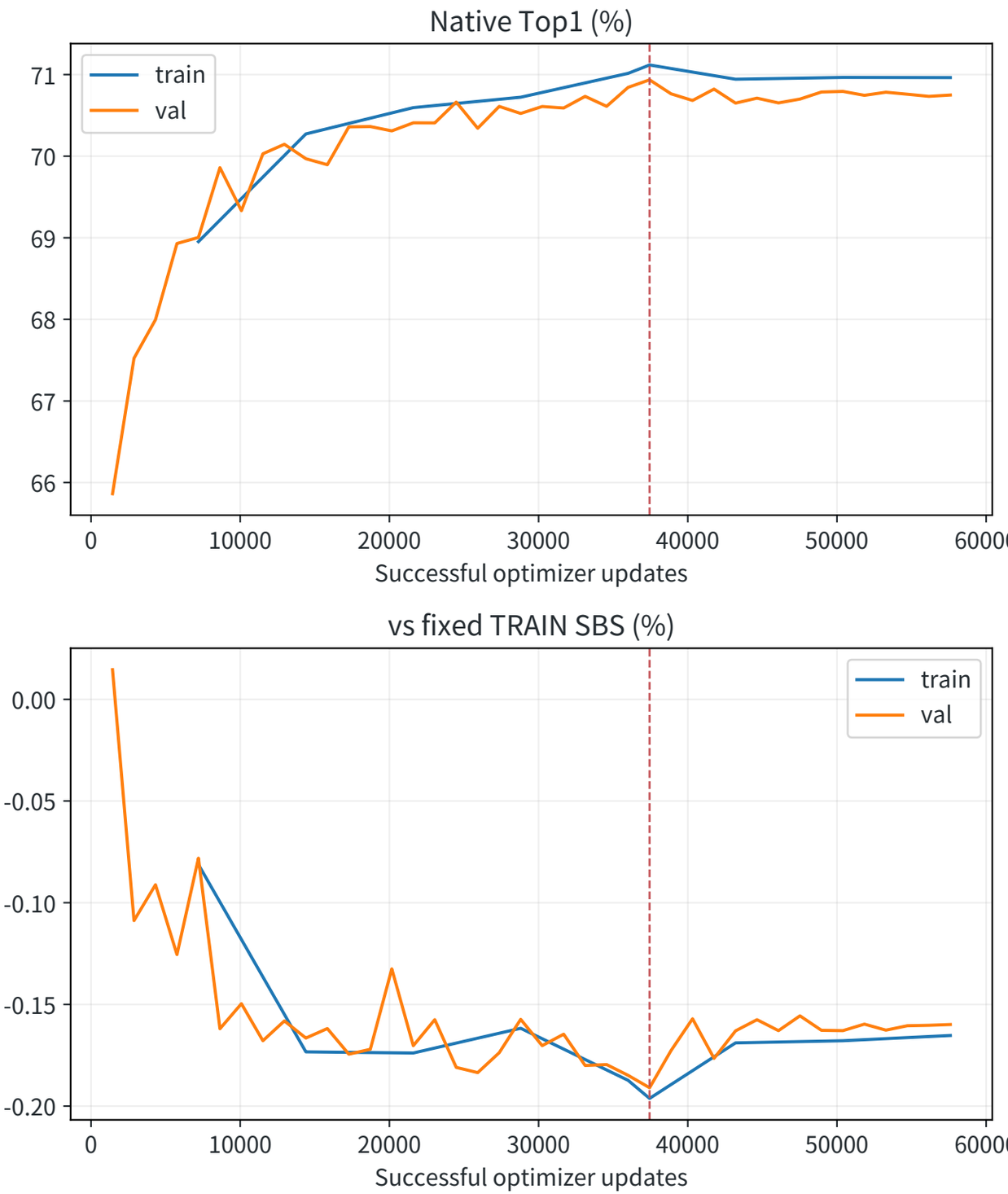R85/B_gradient_balance | successful updates; one locked VAL checkpoint